



Towards climate-smart crops: advances in plant stress biology research

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Abstract

Abiotic and biotic stresses remain major constraints affecting plant growth, development, and crop productivity worldwide. Understanding plant responses to these stresses, both individually and in combination, is therefore essential for developing resilient crop systems and ensuring sustainable agricultural production. This special issue, comprising 32 articles including both review and research papers, highlights recent advances in plant stress biology and provides insights into the mechanisms underlying plant adaptation to diverse environmental challenges. A major focus of the issue is on heat stress tolerance and its interaction with other stresses such as drought, particularly during critical developmental stages in crops such as rice and wheat. Several contributions examine physiological, molecular, and genomic mechanisms associated with stress tolerance, including studies on HSP100 proteins and thermotolerance, drought tolerance in indigenous finger millet, salinity responses in rice and chickpea, and alternative splicing events under salt stress. The issue also includes investigations into plant responses to combined abiotic and biotic stresses, such as heat and aphid stress in wheat, as well as research on plant defense mechanisms against pathogens, including the induction of defense enzymes against mango anthracnose and metabolite-mediated resistance to cucumber downy mildew. In addition, several articles explore beneficial plant–microbe interactions, such as cyanobacterial root microbiomes and microbial consortia that enhance drought tolerance. Advances in genomics and transcriptomics are highlighted through genome-wide analyses of stress-responsive gene families and endophyte-mediated transcriptomic regulation. The issue further covers methodological developments, that integrate artificial intelligence and machine learning with multi-omics data for plant stress research. Collectively, the articles presented in this special issue provide valuable insights into plant stress responses and offer potential strategies for improving crop resilience under changing environmental conditions.

Keywords Abiotic stress · Biotic stress · Crop improvement · Combined stresses · AI-ML in stress biology

Abbreviations

HSP	Heat shock protein
SOS	Salt overly sensitive
MAPK	Mitogen-activated protein kinase pathway
CDPK	Calcium-dependent protein kinase pathway
PPB	Plant-parasitic nematodes (PPN) and plant-pathogenic bacteria
NMD	Nonsense-mediated mRNA decay
GHKL	(Gyrase, H sp90, Histidine Kinase, Mut L)-domain protein

CRISPR/Cas	Clustered regularly interspaced short palindromic repeats/ CRISPR-associated protein
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Introduction

Biotic and abiotic stresses occur both individually and in combination, influencing plant growth, development, and productivity to varying degrees (Pandey et al. 2024). While certain stress interactions may occasionally exert beneficial effects on plant performance, most stress combinations impose negative impacts, leading to significant reductions in crop growth and yield under field conditions (Pandey et al. 2024). Consequently, understanding plant responses to both individual and combined stresses at the mechanistic level

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has become a major focus in plant stress biology (Priya et al. 2023). Approaches such as speed breeding, genome-wide analyses, and detailed investigations of molecular signaling pathways, including mitogen-activated protein kinases and related downstream regulatory genes, have contributed to a deeper understanding of stress response mechanisms. In addition, transcriptomic and other multi-omics approaches are increasingly being employed to unravel complex regulatory networks associated with stress tolerance (Bottini et al. 2025; Rajasekaran et al. 2026a, b). More recently, the application of artificial intelligence and machine learning has emerged as a promising direction for integrating large datasets and identifying key determinants of plant stress resilience.

Plant interactions with biotic agents, including bacteria, fungi, and plant-parasitic nematodes, also play a crucial role in shaping plant health and productivity. Research has highlighted the importance of plant defense mechanisms involving secondary metabolites, induced resistance pathways, and regulatory processes such as nonsense-mediated decay. Metabolite profiling has been widely used to understand these defense responses and their role in stress adaptation. Furthermore, plant responses to combined stresses have become a critical research theme, encompassing interactions between biotic and abiotic stresses, as well as multiple abiotic stress factors occurring simultaneously (Jiang et al. 2025; Chandar et al. 2026). The influence of plant hormones, particularly salicylic acid and related signaling molecules, in modulating tolerance to various stresses has also received considerable attention in recent studies. Collectively, these advances underscore the importance of integrated approaches for understanding plant stress responses and developing strategies to enhance crop resilience (Mu et al. 2025; Zhu and Zhang 2025). Considering these factors, the current special issue titled *Towards Climate-Smart Crops: Advances in Plant Stress Biology Research* presents a collection of articles from these research areas, providing an up-to-date understanding of recent advances in plant stress biology.

Plant responses to abiotic stresses

Crop plants are exposed to several abiotic stresses such as heat, salinity, chilling, and drought, and these stresses trigger diverse physiological, biochemical, and molecular responses (Tiwari et al. 2025; Raja V and Muthamilarasan, 2026; Rajasekaran et al. 2026a, b; Panda et al. 2026). Proline pathway role has been explored in chilling stress (Villasuso et al. 2025). For heat stress, heat shock proteins, particularly members of the HSP family, have been widely investigated for their role in maintaining cellular stability

and thermotolerance (Tripathi and Grover 2025; Fan et al. 2025). In the case of salinity stress, plants activate multiple signaling pathways, including the Salt Overly Sensitive (SOS), Mitogen-Activated Protein Kinase (MAPK), and Calcium-Dependent Protein Kinase (CDPK) pathways, which regulate ion homeostasis and stress-responsive gene expression (Zhang et al. 2025a, b) and alternative splicing (Navaneethakrishnan et al. 2025). Further, relevance of GhSNAT gene family in cotton has been explored (Supriya et al. 2025). In addition, several genomic and molecular tools, including genome-wide analyses and SSR-based molecular markers, are increasingly used in breeding programs to identify stress-tolerant genotypes and accelerate crop improvement. For example, significant variation in physiological traits, ionic balance (Na^+/K^+), and salinity-responsive gene expression among chickpea genotypes at the flowering stage has enabled the identification of potential donors and key traits for breeding salinity-tolerant cultivars (Sathee et al. 2026). Similarly, evaluation of wheat cultivars under high-temperature conditions revealed genotype-specific differences in membrane stability, plant water status, antioxidant responses, and yield-related traits, indicating varied thermotolerance potential useful for breeding programs (Kumar et al. 2026). Recent advances in artificial intelligence further support stress biology research, as AI-driven machine learning and deep learning approaches enable the integration of multi-omics datasets to uncover complex regulatory networks and improve the prediction of stress tolerance traits, thereby accelerating precision breeding of climate-resilient crops. In addition to genetic approaches, biological interventions also contribute to stress mitigation. For instance, combined inoculation of *Azotobacter*, phosphorus-solubilizing bacteria, and *Trichoderma* has been shown to enhance growth, physiological performance, antioxidant activity, and drought tolerance in strawberry plants under water stress conditions.

Plant responses to biotic stresses

Plants are continuously challenged by a wide range of biotic agents including bacteria, fungi, nematodes, and insect pests, which significantly affect crop productivity and quality. Among these interactions, plant-parasitic nematodes (PPN) and plant-pathogenic bacteria (PPB) can interact with host plants to form complex disease syndromes that greatly increase crop losses. Understanding the molecular interactions among PPN, PPB, and host plants, as well as the signaling processes occurring in the rhizosphere, is therefore essential for developing sustainable disease management strategies (Siddiqui and Rizvi 2026). Fungal metabolites also play an important role in plant–microbe interactions.

Secondary metabolites and volatile compounds produced by fungi may act either as virulence factors in pathogenic species or as growth-promoting and defense-inducing agents in beneficial fungi. Understanding the biosynthesis and regulation of these metabolites may enable their exploitation for sustainable agroecological disease control. At the molecular level, the nonsense-mediated mRNA decay (NMD) pathway has emerged as an important regulator of plant immunity by controlling immune receptor expression and preventing autoimmunity (Wang et al. 2012). During pathogen attack, suppression of NMD stabilizes immune-related transcripts, thereby enhancing defense responses, although this may also influence plant growth and development. Several applied studies also demonstrate strategies to enhance plant resistance. For example, treatment with cinnamon oil and 5-sulfosalicylic acid induces defense-related enzymes in mango fruits against *Colletotrichum gloeosporioides*, indicating an eco-friendly approach for postharvest disease management through activation of phenylpropanoid-mediated resistance. Similarly, a ~46 kDa GHKL-domain protein from *Bacillus subtilis* has shown strong insecticidal activity against *Helicoverpa armigera* larvae, highlighting its potential as a biological control agent for sustainable pest management in pigeonpea (Singh et al. 2026a, b). In addition, metabolite profiling using GC-MS and molecular docking analysis suggests that intrinsic phytochemicals present in the downy mildew-resistant cucumber line PI-197,087 may inhibit pathogen effectors, thereby contributing to natural biochemical resistance mechanisms. Also, the induced systemic resistance in mango fruits against anthracnose salicylic acid analogue and other antagonist have been explored (Pandita et al. 2026).

Plant responses to combined stresses

Climate change alters temperature, precipitation patterns, and CO₂ levels, which can intensify pathogen spread and plant disease severity (Farda et al. 2026; Choudhary and Senthil-Kumar 2024). Integrated approaches including resistant varieties, precision agriculture, and improved management practices are required to sustain crop health (Rai 2022). Reproductive-stage drought and heat stress in rice, particularly under combined conditions, severely disrupt floral development, reduce pollen viability, impair stigma receptivity, and result in significant yield losses, highlighting reproductive structures as important targets for breeding stress-tolerant varieties (Nahakpam et al. 2026; Zhu 2025). Similarly, evaluation of wheat genotypes under simultaneous heat and aphid stress showed that combined stress intensifies yield losses and induces stronger biochemical responses (Reddy et al. 2026), while genotypes HD2967

and DBW333 displayed relatively better performance and useful traits for stress resilience. Hormonal regulation also contributes to stress mitigation; for example, exogenous salicylic acid alleviates chromium-induced oxidative stress in pea by enhancing antioxidant systems, regulating glutathione metabolism, and reducing metal translocation (Singh et al. 2026a, b). In addition, genome-editing technologies such as CRISPR/Cas, along with base and prime editing approaches, enable precise genome modifications and are increasingly used to enhance yield, stress tolerance, and disease resistance in crops (Chilakala et al. 2023; Alptekin 2025).

Conclusions and perspectives

This special issue, containing the above-mentioned contributions, provides recent updates on plant responses to abiotic stresses, biotic stresses, and their combined interactions under natural conditions. It covers physiological, biochemical, and molecular aspects studied across multiple crops, including millets, chickpea, legumes, and several other plant species. The articles also highlight the use of various tools, including molecular breeding approaches and artificial intelligence-based methods, thereby offering both mechanistic insights and application-oriented perspectives. In future, research in this area may move further toward the application of microbial consortia for mitigating both biotic and abiotic stresses, as well as the use of artificial intelligence and machine learning tools at field level and for analysis of large datasets such as omics data and predictive models. In addition, CRISPR-based genome editing approaches, as indicated in the articles included in this issue, are expected to play an important role in advancing research and crop improvement in plant stress biology.

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